ANGRY RAINS

Climate changed has altered rainfall patterns—monsoons now marked by short, intense spells

EFFECT

- Torrential rain from the cloudburst in early October fell on deforested slopes and urbanised plains, overwhelming the landscape
- Critical infrastructure and economy damaged in the floods include highways, the Darjeeling Himalayan Railway, tea estates and croplands

CLOGGED RIVERS

N. Bengal rivers have lost old channels, wetlands and floodplains to agriculture & urbanisation

- **Torsa and Raidak:** Lost old channels to farmland
- **Kaljani, Jaldhaka:** Heavily embanked
- **Teesta:** Lost channels connecting it to wetlands to habitations

EFFECT

- When torrential rain fell on October 5-6, these river basins could not absorb water like before, causing floods





mountainsides are composed largely of shale, sandstone and metasediments, which are inherently unstable. Annual rainfall between 750 mm and 1,000 mm make them among India's most landslide-prone areas. Data from the Indian Space Research Organisation's (ISRO) Landslide Atlas of India (2023) shows 172 mapped landslides in Bengal between 1998 and 2022, almost all in these two districts. Hundreds more go unmonitored. Slope failures here evolve into debris flows that tear through roads, tea gardens and settlements. Population density and infrastructure on fragile slopes heighten vulnerability. Rainfall, of course, is the primary trigger for landslides and floods. Driven by climate change, the monsoon is now marked by short, intense cloudbursts. When torrential rain falls on deforested slopes and urbanised plains, the land's ability to absorb water collapses.

FUTURE COURSE OF ACTION

Experts argue that the only road forward is to restore the landscape's natural resilience. Rivers must be reconnected to their floodplains. Encroachment must be halted, and land use regulated. Urban planning in Darjeeling and Kalimpong towns must integrate landslide susceptibility maps. Emerging technologies offer new tools: ISRO's Persistent Scatterer InSAR (PSInSAR) can detect slope movements at the millimetre scale, providing early warnings of landslides. Remote sensing can map land-use changes over time, guiding targeted interventions.

"Intense rainfall, reckless development and institutional neglect are damaging assets like hydropower projects, highways and military infrastructure. India must create Himalayan-focused institutions for disaster management," says Prof. Mahendra P. Lama, former member of the National Security Advisory Board. He supports the demand for an Indo-Bhutan river commission.

The October 5 floods serve as a warning—unless rivers, forests, hills and settlements are brought back into ecological alignment, a seasonal challenge will become a chronic disaster. ■